Milan Barot

Healthcare Payer Data Engineer | Claims, Eligibility, EDI 834/837, HRP | SQL Server, SSIS, ADF, Databricks
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Summary

Healthcare data engineer with 6 years of experience, including nearly 4 years in US healthcare payer/TPA data at Blue Cross Blue Shield of Arizona and AmeriBen. Builds and supports SQL Server, SSIS, and Azure Data Factory pipelines for claims, eligibility, accumulator, provider, value-based care, and financial data, including EDI 834/837 file delivery to CMS and vendors such as OptumRx and HealthEquity. Hands-on with HealthRules Payer (HRP) migrations, reconciling legacy and HRP-derived data, and supporting mission-critical CMS production releases, using Devin and Windsurf AI coding assistants with human review in day-to-day development. Currently expanding into Databricks, Apache Spark, and Delta Lake lakehouse architectures.

Technical Skills

Healthcare Domain: Claims, eligibility & enrollment, accumulators, provider, value-based care (VBC), EDI 834/837, CMS submissions, HealthRules Payer (HRP), HIPAA
Data Engineering: SQL, Python, PySpark, Apache Spark, SSIS, Azure Data Factory, ETL
Cloud & Lakehouse: Azure, Databricks, Delta Lake, Medallion Architecture
Databases/Data Warehouse: SQL Server, Azure SQL, PostgreSQL, DuckDB
BI & Reporting: Power BI, SSRS
Data Integration: EDI 834/837, REST APIs, FTP, ActiveBatch, SQL Server Agent
Workflow & DevOps: Airflow, Dagster, Docker, Git, Jenkins, Jira
Application & AI: Devin, Windsurf, FastAPI, React, ChromaDB, OpenAI API, Groq API

Work Experience

SQL Server Developer / Data Engineer Jan 2026 – Present
AmeriBen (UST Global — Implementation Partner) *Remote*

- Support AmeriBen's migration of healthcare claims reporting from the legacy **GBAS** platform to **HealthRules Payer (HRP)**, keeping financial, claims, utilization, census, and clinical reports accurate for self-funded employer groups as roughly 20 client groups move to the new platform.
- Built merged reports that draw from both GBAS and HRP for clients split across the two systems, including **stop-loss rolling summary** and **large claims** reports, so each client keeps one consistent view of its plan data during the transition.
- Migrate and maintain reports across two report families (about 10 AmeriBen Reporting Suite reports and 10–15 Reporting ODS reports) and am helping deploy **roughly 200 more reports** ahead of a January 2027 deadline.
- Use **Devin** and **Windsurf** AI coding assistants to speed up SQL development, debugging, and report rebuilds, and validate AI-rebuilt reports against existing production logic before they are deployed.
- Resolve production incidents and calculation defects in migrated claims data, such as lag calculations on adjusted claims, by tracing issues through the HPS → ODS pipeline to the source instead of patching the report layer.
- Develop and maintain the **SQL Server** stored procedures (some over 2,000 lines), merge scripts, and **ETL** workflows behind large claims, financial summary, contract savings, utilization, lag analysis, and level-of-coverage census reports.
- Work with business stakeholders and HRP teams to turn payer/TPA reporting requirements into validated data mappings and business rules that meet migration milestones.
- **Tech Stack:** SQL Server 2022, Azure SQL, SSRS, SSIS, ETL, HealthRules Payer (HRP), Devin, Windsurf, TFS.

Data Engineer Sep 2022 – Oct 2025
Blue Cross Blue Shield Arizona (Mastek — Implementation Partner) *Phoenix, AZ*

- Developed a consolidated **EDI framework for 834 and 837** enrollments, claims and dental, using three core SSIS packages to automate the generation and delivery of over **35 distinct file types** to CMS. This architecture ensured secure data exchange and audit readiness while significantly reducing the manual maintenance and operational overhead previously required.
- Provided active production support for mission-critical CMS data exchanges, conducting over **25 successful production deployments** and managing emergency releases while maintaining compliance with federal reporting deadlines.

- Designed and implemented end-to-end **ETL** pipelines using **SSIS** and **SQL Server** to process eligibility, claims, accumulator, and value-based care (VBC) files, ensuring compliance with business requirements and vendor-specific standards. Collaborated with vendors such as OptumRx, HealthEquity, Equality Health, and others to deliver high-quality data solutions.
- Utilized **Azure Data Factory (ADF)** to design and orchestrate scalable data integration pipelines for migrating legacy member and claims data to **HealthRules Payer (HRP)** on Azure Cloud.
- Rebuilt critical ETL pipelines using SSIS and SQL Server after the **CHC system hack (2024)**, restoring high-priority data workflows under tight deadlines with minimal downtime.
- Led the rapid resolution of critical production failures, ensuring minimal data loss during high-volume enrollment periods where direct connectivity with CMS was required.
- Collaborated closely with **Deloitte** on the HealthEquity project to develop eligibility and claims files and automated it end to end. Responsibilities included gathering business requirements, creating technical documentation, optimizing ETL pipelines, conducting performance testing, and ensuring smooth deployment into production environments.
- Leveraged **Azure SQL Database** for staging, transformation, and persistent storage of high-volume healthcare data. Configured ADF linked services, pipelines, and triggers for automated, scheduled data ingestion from on-premises SQL Server to Azure.
- Developed complex data extracts that combined on-prem warehouse data with select cloud-based sources, ensuring consistency, accuracy, and seamless integration across hybrid environments, reducing execution time by **25%** on average across the board.
- Automated data workflows by integrating SQL Server Agent jobs with **ActiveBatch**, setting up FTP connectivity with strict naming conventions and cadence-driven schedules for file delivery to vendors.
- Led offshore engineering resources across multiple ETL initiatives, managing task assignments, technical dependencies, and delivery timelines.
- Partnered with business stakeholders to gather requirements for new projects, conducting detailed assessments of complexity, technical effort, development timelines, and warranty periods. Worked with Mastek to draft SOWs for approval by Blue Cross Blue Shield, ensuring alignment with business objectives.
- **Tech Stack:** Azure Data Factory (ADF), Azure SQL Database, C#, SQL Server, SSIS, SSRS, ActiveBatch, SQL Server Agent, FTP, EDI (834, 837), HealthRules Payer (HRP), Agile.

Data Engineer Feb 2021 – Sep 2022

CBRE *Dallas, TX*

- Designed, developed, and maintained robust ETL pipelines using **Python, SQL, and PostgreSQL** via CBRE's internal ETL framework (Lymbic). Translated complex business logic from real estate analysts into SQL/Python transformations for property usage, tenant risk, and leasing performance.
- Ingested diverse external data sources such as Google Mobility Reports and BREAM energy efficiency assessments, transforming them into structured datasets in Parquet and CSV formats that fed Power BI dashboards used by leadership during the COVID-19 pandemic.
- Applied data profiling and transformation using **Pandas** and **PySpark** for efficient in-memory processing of large CSV/Parquet files, enabling faster QA cycles and stakeholder feedback loops.
- Supported **Databricks** orchestration workflows — monitoring job execution, rerunning failed tasks, and assisting engineers with job recovery — ensuring daily and weekly batch pipelines remained consistent and reliable.
- **Tech Stack:** Python, SQL, PostgreSQL, Power BI, Airflow, Pandas, Parquet, Agile.

ETL / SQL Server Developer Oct 2019 – Aug 2020

AmNet Services Inc. *Fairfield, NJ*

- Developed **SSIS** packages to automate extraction and ingestion of network test data from Excel, flat files, and MS Access sources into a SQL Server staging environment.
- Designed and implemented a dimensional data warehouse model using **ERwin Data Modeler** to support analytical queries on network KPIs.
- Wrote complex **T-SQL** stored procedures for data cleansing, aggregation, and validation, and built **SSRS** reports and Power BI dashboards for signal strength, call drop rates, and E911 test call success metrics.
- **Tech Stack:** SSIS/SSRS, T-SQL, Power BI, Agile.

Selected Projects

Kalshi-Prediction-Engine — AI-Powered Intelligence Lake-House | *PySpark, Delta Lake, ChromaDB, Groq, React*

- Architected a full-stack prediction market intelligence platform using a **Medallion (Bronze/Silver/Gold) data lakehouse**, processing 6 global news feeds, GDELT event streams, and Kalshi prediction market data through **PySpark ETL** with **Delta Lake ACID** guarantees.
- Engineered a **RAG-powered inference engine** that embeds Silver-layer data into ChromaDB (MiniLM-L6-v2), performs cascading time-windowed semantic search, and synthesizes intelligence briefs via Groq (Llama 3.3 70B).
- Deployed a **10-container Docker Compose** production environment with a custom ETL orchestrator (watermark-based change detection, lock-file API coordination, once-daily settlement sweeps) running autonomously at $0/month on local infrastructure.

Education

Washington University of Science & Technology — Vienna, VA
B.S. in Information Technology — Software Engineering — *Mar 2022*

The City University of New York — New York, NY
B.A.S. in Computer Information Systems — *Dec 2019*

Certifications

Databricks Certified Data Engineer Associate — Nov 2026